

EE5203 Lab 3 Guide

Functions and registers: a sensor module and an LED driver for LD2 - Basri Erdogan

Mission: Move your Lab 2 calculations into C functions, and write a small LED driver that controls LD2 (pin PA5) through CMSIS registers. At the end, LD2 blinks the status of your eight sensor values: **1 blink = LOW, 2 blinks = OK, 3 blinks = HIGH**.

This lab uses Lecture 3 (functions, header files, `CMakeLists.txt`) and Lecture 4 (registers, bit masks, `->`).

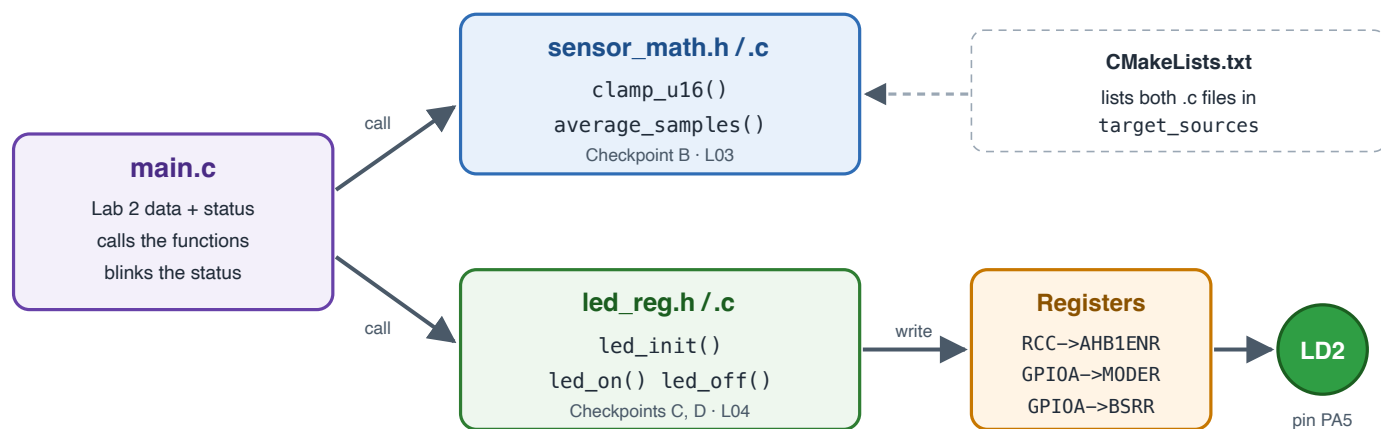


Figure 1: What you build: `main.c` calls two small modules; the LED module writes three registers that drive LD2.

Prepare before class

- ☐ Bring your board and a USB data cable, and your **Lab 2 results** (the Checkpoint D table). Lab 3 must give the same values.
- ☐ Download the starter files: [EE5203_L03_starter.zip](#).
- ☐ Do the **Lab 3 pre-lab quiz** on CATS. It includes the bit-mask table below. To practise masks first, use the [Bitwise Lab](#).
- ☐ Bring this table, completed on paper:

Purpose	Expression	Hex value
GPIOA clock bit	<code>1U << 0</code>	
PA5 mode field (two bits)	<code>3U << 10</code>	
PA5 output pattern 01	<code>1U << 10</code>	
Set PA5	<code>1U << 5</code>	
Reset PA5	<code>1U << 21</code>	

Learning objectives

By checkoff time, you can:

- write a function declaration in a `.h` file and its definition in a `.c` file;
- add a new `.c` file to the build in `CMakeLists.txt`;
- replace repeated code in `main.c` with function calls and get the same results;
- set and clear register bits with `|=`, `&=` and masks;
- show in the debugger that a register value agrees with what LD2 does.

Hardware and files

Item	Value
Board	Nucleo-F401RE
Tools	STM32CubeMX (new project) + VS Code
Starting point	A new CubeMX project; the Lab 2 code is given in Checkpoint B
Project name	<code>EE5203_L03_<studentnumber></code>
New files (starter)	<code>sensor_math.h</code> , <code>sensor_math.c</code> , <code>led_reg.h</code> , <code>led_reg.c</code>
Files you edit	<code>CMakeLists.txt</code> , <code>main.c</code> , and the four new files
Registers	<code>RCC->AHB1ENR</code> , <code>GPIOA->MODER</code> , <code>GPIOA->BSRR</code>

Keep all edits to `main.c` inside the `USER CODE` regions.

Checkpoint A - Create a new project and add the starter files

Start of the session: if `C:\Work\<studentnumber>` does not contain your Lab 2 project, copy **all** your project folders from `H:\EE_Labs` into it first (as in Lab 2). Do not run *Save to H* now.

You make a **new** project. Do not change your Lab 2 project.

1. In **STM32CubeMX**: Board Selector → **NUCLEO-F401RE** → accept the default initialization (as in Lab 1). PA5 is already LD2.
2. Project Manager: name **EE5203_L03_<studentnumber>**, location `C:\Work\<studentnumber>`, **Toolchain/IDE = CMake** → Generate Code.
3. Copy the folder `C:\Users\Public\EE5203_template\.vscode` into the new project folder, next to the `.ioc` file.
4. Open `EE5203_L03_starter.zip`. Copy the **Core** folder inside it and paste it into your project folder. Two files go to `Core\Inc`, two to `Core\Src`.
5. In `CMakeLists.txt`, add the two new `.c` files to `target_sources`, below the line `# Add user sources here`. Write the paths yourself, in the same form as `Core/Src/<file>.c`. (The project name is already correct: CubeMX used your folder name.)
6. Open the folder in VS Code and build with **Ctrl+Shift+B**: zero errors. The starter files build even though they are not finished. Two warnings (`unused parameter 'low'` and `'high'`) are normal: they go away when you finish `clamp_u16` in Checkpoint B.

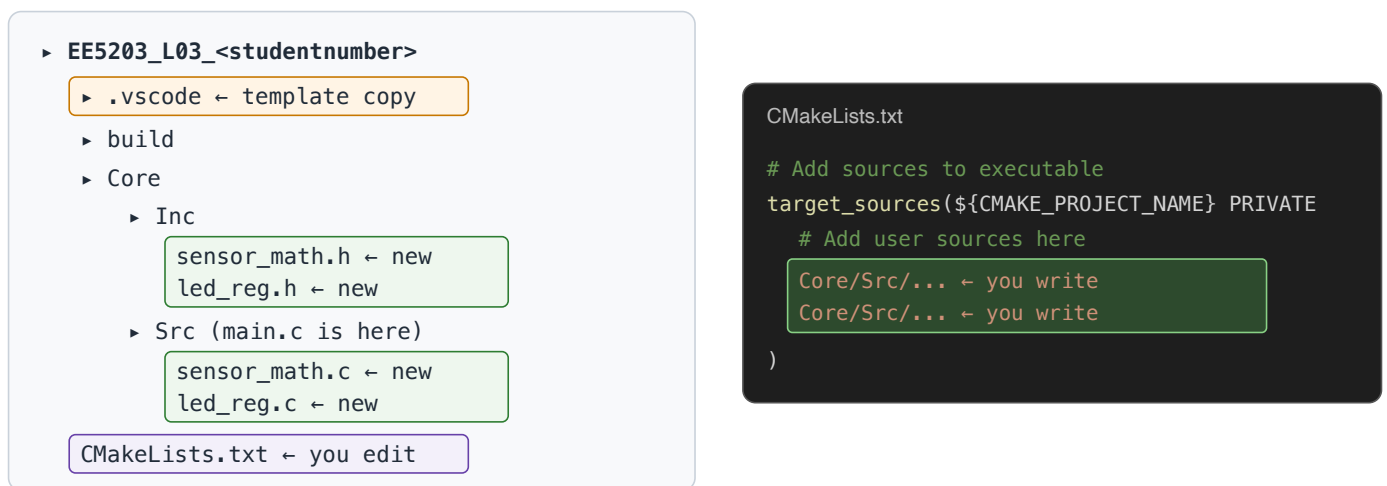


Figure 2: Your project folder after Checkpoint A.

Show: the four new files in the Explorer, your two lines in `CMakeLists.txt`, and the build line `Linking C executable EE5203_L03_<studentnumber>.elf`.

Checkpoint B - Move the Lab 2 calculations into functions

You write two short functions, then call them from `main.c`.

B1 - Write the two functions

1. In `sensor_math.c`, complete the **body** of `clamp_u16`: return `low` if the value is below `low`, `high` if it is above `high`, and the value itself otherwise.
2. In `sensor_math.h`, replace the `TODO` comment for `average_samples` with this **declaration**:

```
uint16_t average_samples(uint16_t values[], int count);
```

3. In `sensor_math.c`, replace the `TODO` comment for `average_samples` with this **definition**, and write the loop yourself:

```
uint16_t average_samples(uint16_t values[], int count)
{
    uint32_t sum = 0;
    /* Your loop: add every values[i] to sum. Use i < count. */

    return sum / count;
}
```

4. Build: zero errors.

B2 - Call the functions from main.c

1. Add `#include "sensor_math.h"` inside USER CODE BEGIN Includes.

Add steps 2-4 inside USER CODE BEGIN 2, one below the other.

2. The eight values from Lab 2:

```
uint16_t samples[8] = {850, 1250, 4095, 4200, 2050, 900, 3100, 2800};
```

3. Clamp every value with your function:

```
for (int i = 0; i < 8; i++) {  
    samples[i] = clamp_u16(samples[i], 0, 4095);  
}
```

4. Calculate the average with your function:

```
uint16_t average = average_samples(samples, 8);
```

5. Build and start the debugger.

The results must be **the same as in Lab 2**. Copy the Lab 2 values from your Lab 2 table, then read the Lab 3 values in the debugger:

Quantity	Lab 2 value	Lab 3 value (debugger)
<code>samples[3]</code> after the loop		
<code>average</code>		

Show: a breakpoint inside the loop of `average_samples`, with `i` and the sum changing as you step, and the completed table.

Checkpoint C - Write `led_init()` and check the registers

`led_reg.h` and `led_reg.c` contain only comments. You write the code. The three registers you use in Checkpoints C and D:

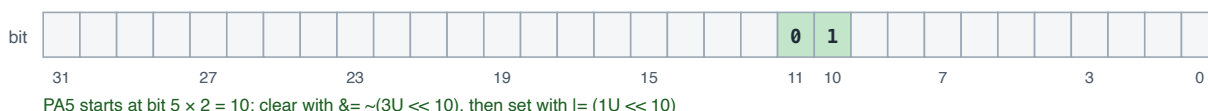
RCC->AHB1ENR

bit 0 = GPIOA clock on · use `1=` (keep the other bits)



GPIOA->MODER

bits 11:10 = PA5 mode · 00 input, 01 output



GPIOA->BSRR

write-only · use = (a 1 acts, a 0 does nothing)

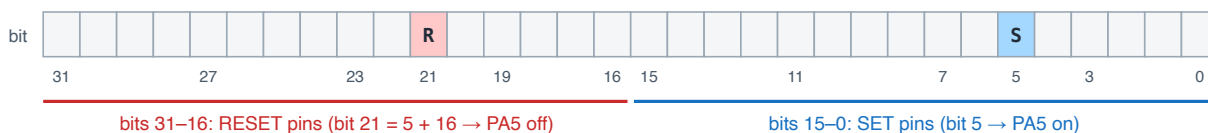


Figure 3: The bits you change: AHB1ENR bit 0, MODER bits 11:10 for PA5, BSRR bit 5 (set) and bit 21 (reset).

1. In `led_reg.h`, **declare** three functions with no parameters and no return value: `led_init`, `led_on`, `led_off`.
2. In `led_reg.c`, **define** `led_init` with three statements:
 - set bit 0 of `RCC->AHB1ENR` (GPIOA clock) without changing other bits;
 - clear bits 11:10 of `GPIOA->MODER`;
 - write the pattern 01 into bits 11:10 (PA5 = output).
3. In `main.c`, add `#include "led_reg.h"` inside USER CODE BEGIN Includes.
4. Call `led_init()`; inside USER CODE BEGIN `SysInit`. This region is just **above** `MX_GPIO_Init()`; so your function runs **before** the CubeMX code:

```
/* USER CODE BEGIN SysInit */
led_init();
/* USER CODE END SysInit */
```

At this point nothing has touched GPIOA yet: the clock bit is 0 and PA5 is in input mode (00). Your three statements change both.

Break on `led_init()`; then step **into** it (F11) and over each statement. Look at GPIOA and RCC in the register (SFR) view:

Moment	RCC->AHB1ENR bit 0	MODER bits 11:10
Before <code>led_init</code>		
After the clock line		
After the clear		
After the set		

Then step over `MX_GPIO_Init()`; CubeMX writes the same values, so nothing changes.

Show: the table, and answer: if the clock bit is 0, what happens to a write to `GPIOA->MODER`?

Checkpoint D - Turn LD2 on and off with BSRR

- In `led_reg.c`, define `led_on` and `led_off`. Each is one assignment to `GPIOA->BSRR`: bit 5 sets PA5, bit 21 (5 + 16) resets it.
- In `main.c`, inside `USER CODE BEGIN 3`, add this blink (LD2 on, wait, off, wait; the `while` (1) loop repeats it):

```
led_on();
HAL_Delay(200);
led_off();
HAL_Delay(200);
```

Do not use any `HAL_GPIO_...` call.

- Put a breakpoint on `led_off()`; When it stops, LD2 must be **on**. Step over: LD2 turns **off**.

Show: LD2 blinking, your three functions, and the register view when the program stops at `led_off()`; (ODR bit 5 = 1).

Checkpoint E (optional, not graded) - Blink the status

First add the Lab 2 status rule in `USER CODE BEGIN 2`, below the `average` line:

```
int status_code = 0;
if (average < 1500) {
    status_code = -1;    /* LOW */
} else if (average > 3000) {
    status_code = 1;    /* HIGH */
}
/* otherwise 0 = OK */
```



LD2 on = line up, LD2 off = line down.

Figure 4: The three blink patterns.

- Calculate a blink count: LOW = 1, OK = 2, HIGH = 3. One short expression is enough.
- In `USER CODE BEGIN 3`, replace the Checkpoint D blink: blink LD2 that many times (200 ms on, 200 ms off), then wait 1000 ms. Use a `for` loop.

3. Predict the number of blinks for your data before you run it.

Show: the blink pattern, your prediction, and the code.

If you finish early: change one of your samples so that the status becomes HIGH, predict the new blink count, and check it.

Common problems

What you see	First check
undefined reference to 'average_samples' or 'led_init'	The .c file is in target_sources . The name in the .c file matches the .h file.
implicit declaration of function 'led_on'	#include "led_reg.h" is in main.c , and the declaration is in the header.
'GPIOA' undeclared in led_reg.c	#include "main.h" is at the top of led_reg.c .
conflicting types for 'average_samples'	The declaration and the definition use exactly the same types.
LD2 never lights after led_init	Check the MODER bits in the register view: they must be 01 .
MODER stays 00 while you step through led_init	The clock line must come first . Without the clock, writes to GPIOA are lost.
LD2 stays on	led_off uses bit 21, not bit 5.
The average differs from Lab 2	Stop inside average_samples ; watch i , count and the sum.
...FLASH.ld: syntax error	Build with Ctrl+Shift+B (the template repairs the CubeMX linker script). Check that .vscode is in the project folder.

End of the session - Save to H

Close VS Code, run the **Save to H** desktop shortcut, and check that **H:\EE_Labs\EE5203_L03_<studentnumber>** exists before you log out.

Acceptance checklist

- ☐ The build is clean. Every declaration matches its definition.
- ☐ Lab 3 results equal the Lab 2 results (Checkpoint B table), and the Lab 2 project is unchanged.
- ☐ **led_init**, **led_on**, **led_off** use only register writes; there is no **HAL_GPIO_...** call in your code.
- ☐ The register table of Checkpoint C is filled in.
- ☐ LD2 blinks with **led_on** / **led_off** (Checkpoint D).
- ☐ All **main.c** edits are inside the **USER CODE** regions.

Individual oral check

Be ready for **one** question **or** one live change, chosen by your lab assistant. For a live change, predict the result first, then make the change.

Questions:

- What is the difference between a declaration and a definition? Where is each one for **led_on**?
- Why does **average_samples** need the count as a parameter?
- Why does **led_init** use **!=** for the clock, but **led_on** uses **=** for **BSRR**?
- What does **3U << 10** select, and why does **PA5** start at bit 10?

Live changes:

- Change both blink delays to 100 ms. Do the masks change?
- Change the **HIGH** limit from 3000 to 2000. How many blinks now?
- Replace the literal 5 with **#define LED_PIN 5U** in **led_on** and **led_off**.

Submit

Submit the four module files, **CMakeLists.txt**, the **main.c** user-code sections, the Checkpoint B and C tables, and one photo or short video of LD2 blinking.

Marks (10): pre-lab quiz on CATS 1 · A new project and build 1 · B functions 2 · C **led_init** and registers 2 · D **led_on/led_off** 2 · oral check (one question or one live change) 2. Checkpoint E is optional and not graded.